University of Applied Sciences
Hochschule
Kufstein Tirol

FH KUFSTEIN TIROL I DSNSE WS 2026/27

Pump Failure: Early Predictive Maintenance

Strategic Fleet Audit, Governance & Economic Modeling (Phase 0)

Lead Data Scientist: Ian Calixto
Program: MSc Data Science & Intelligent Analytics
Presentation Date: November 06, 2026



01 The Industrial Setting & Operational Reality

Fleet Operations, Production Costs, and Maintenance Bottlenecks

University of Applied Sciences
Hochschule Kufstein Tirol

200 Pumps
Centrifugal Fleet
Sites S1, S2, and S3

52 Weeks
Telemetry History
10,209 valid weeks (2025)

12 Pumps / wk
Sunday Capacity Limit
1 technician capacity

€18,000
Catastrophic Failure Cost
vs €1,200 proactive inspection

The Operational Sunday-Evening Mandate:

- Every Sunday at 23:59, the plant manager evaluates all 200 pumps.
- Generates **"The Sunday List"**: a prioritized schedule of at most **12 pumps**.
- Technicians execute preventive maintenance on Monday morning.
- Goal: Maximize catastrophic failures prevented within the strict 12-visit limit.

Asymmetric Economic Threshold:

$$\text{Threshold}^* = \frac{C_{\text{inspect}}}{C_{\text{fail}}} = \frac{€1.200}{€18.000} \approx \mathbf{6.67\%}$$

- **Decision Rule:** Any pump exceeding **6.67% failure risk** economically justifies technician inspection!
- Symmetric 0.5 threshold fails completely under 15:1 cost asymmetry.

02 Empirical Data Audit & Two-Tier Pipeline

Eliminating SCADA Glitches, Hardware Gaps, and Sensor Dropouts

University of Applied Sciences
Hochschule Kufstein Tirol

Empirical Finding	Volume	Engineering Fix (DE-05)
`/record_id` slash	Header	Cleaned in `rename_record_id()`
SCADA Duplicates	40 pairs (80 rows)	`drop_duplicates(pump_id, week)`
Commissioning Gaps	191 weeks (9 pumps)	Retain active weeks (no fake rows)
Hardware Sensor Absence	25 vib / 29 RTD	Flags: `has_vibration_sensor`
Sensor 0.0 Dropouts	94 operating rows	Replaced `vibration == 0` with `NaN`

Two-Tier Architecture (Zero Leakage):

Tier 1 (Clean Pipeline - `src/pumpfail/clean.py`):
Deterministic row-level sanitation (IDs, duplicates, sensor dropouts to NaN). Exported to `data/interim/pump_clean.csv` (10,209 x 32).

Tier 2 (CV Estimator Pipelines):
Statistical imputation (median/KNN) and scaling strictly delayed into cross-validation folds. **Zero fold leakage!**

03 Feature Governance & Quarantining Target Leakage

The Sunday-Evening Deployment Condition & Explicit Guardrails

University of Applied Sciences
Hochschule Kufstein Tirol

26 Features
Legitimate Predictive Predictors
20 numeric, 4 categorical, 2 flags

1 Target
Ground Truth Target
`failure_next_week` (binary 0/1)

2 Quarantined
Strict Leakage / Diagnostic
`repair_cost_eur`, `failure_mode`

The Fatal Target Leak: `repair_cost_eur`

- Exactly **402 non-null values** corresponding 100% to the **402 failures**.
- Invoices only exist **post-mortem** after emergency repair teardown!
- At Sunday 23:59, the future repair invoice does not exist.
- Training on it creates a 100% fake “perfect” model useless in production.

Programmatic Guardrails:

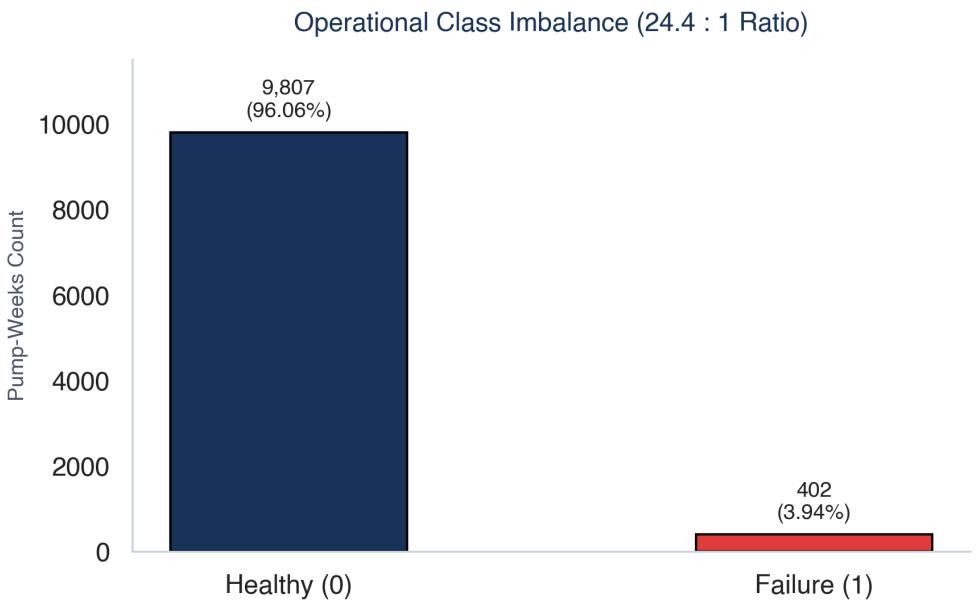
- `src/pumpfail/features.py` enforces:

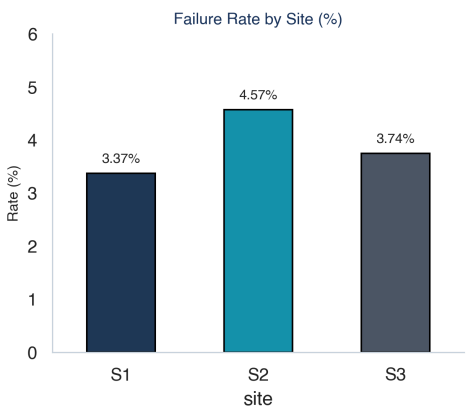
```
validate_no_leakage(features)
split_features_target(df)
```

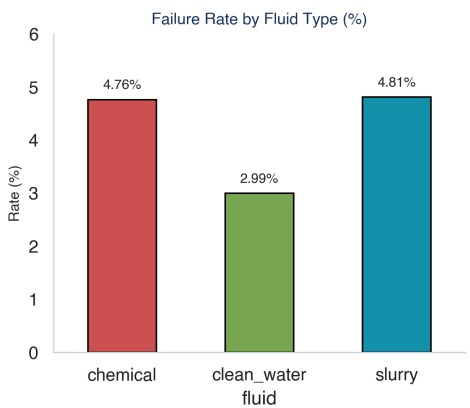
- Raises `LeakageError` immediately if post-incident variables enter X.

04 Fleet Failure Dynamics & Severe Imbalance

Rare-Event Characteristics Across 10,209 Operating Weeks





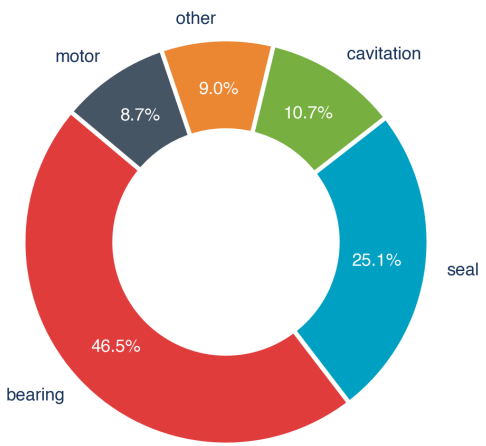


Key Takeaways: Fleet baseline failure rate is **3.94%** (1 failure per 25 operating weeks). Plant **S2** has the highest failure rate (**4.57%**). Fluid dynamics match engineering physics: abrasive **slurry (4.81%)** and corrosive **chemical (4.76%)** cause 60% more failures than **clean water (2.99%)**.

05 Failure Modes & Statistical Information Capacity

Effective Sample Size vs Serial Entity Autocorrelation

Root Causes of 402 Failures (Bearings + Seals = 71.6%)

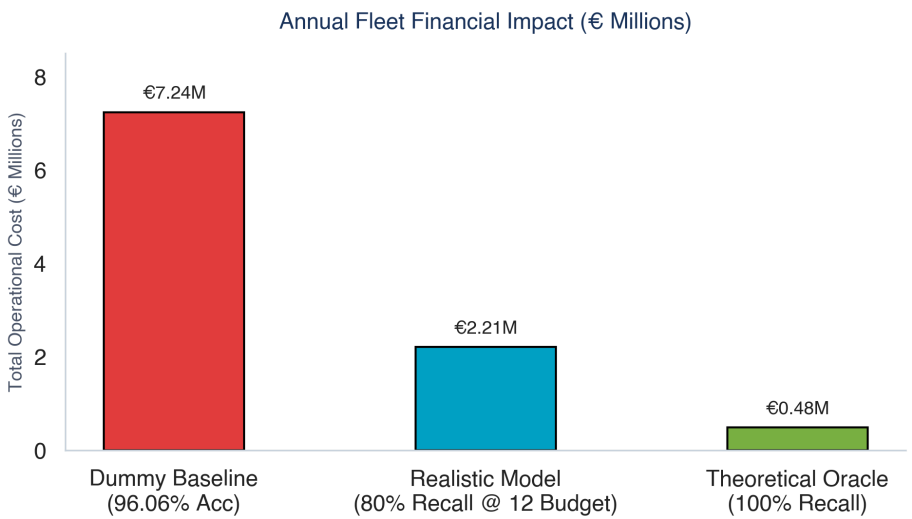


Statistical Information Limits ($N_{\text{eff}} \ll 10.209$):

- **Mechanical Wear Dominates:** Bearings (46.5%) and Seals (25.1%) account for **71.6% of all fleet failures**. Motor coil burnout is rare (8.7%).
- **15 Pumps (7.5%) Never Fail:** Operate 52 weeks in pristine condition; provide healthy baselines but zero pre-failure degradation signatures.
- **Intra-Pump Autocorrelation:** 52 weeks per pump are not $i.i.d.$ observations. Random splits leak pump baseline identities!
- **Effective Sample Size:** We only observe **402 failure transitions**.

06 The Intern's Fallacy: The €7.24M Catastrophe

Why 96.06% Accuracy Means 100% Failure to the Business



The 3 Executive Claims to Management:

1. **Claim A (The Deceptive Accuracy Trap):** Predicting that zero pumps will ever fail automatically yields **96.06% accuracy** without learning any operational signal.
2. **Claim B (The Operational Disaster):** At default threshold 0.5, the model detected **0 failures (Recall = 0.0%)**, sending zero technicians and causing **€7,236,000 in unmitigated outages**!
3. **The Resolution:** A cost-calibrated model ($\tau^* \approx 6.67\%$) operating under the 12-inspection Sunday budget captures the catastrophic breakdowns, unlocking up to **€6.75M in annual savings**!

07 Phase 1 Engineering & Modeling Roadmap

Leakage-Free Validation, Cost Matrix, and The Sunday List

University of Applied Sciences
Hochschule
Kufstein Tirol

1. GroupKFold Validation

- Grouped by `pump_id`.
- Guarantees zero pumps appear simultaneously in train and test folds.
- Evaluates generalization to unobserved machinery.

2. Temporal Holdout Split

- Weeks 1–39: Cross-validation training and threshold tuning.
- Weeks 40–52 (Q4): Pure temporal holdout evaluated exactly once.
- Prevents future look-ahead bias.

3. Capacity Optimization

- Optimize Precision-Recall curve.
- Sunday Top-12 ranking ($\mathrm{Recall@12}$).
- Financial cost minimization table directly in Euros (€).

Thank You! | Questions & Technical Defense

FH Kufstein Tirol – Data Science & Intelligent Analytics (WS 2026/27)